

Artificial Intelligence Basics Course Summary

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1 Key Terms

- **Minimax Search:** A search algorithm for two-player zero-sum games that aims to find the best move for the current player, assuming the opponent also plays optimally.
- **Alpha{-}Beta Pruning:** A technique to reduce the search space in Minimax search by avoiding the evaluation of branches that cannot affect the optimal decision.
- **Turing Test:** An operational test for intelligent behavior, proposed by Alan Turing, where a human evaluator must distinguish between a human and a machine based on their textual responses.
- **Game Tree:** A tree-like structure representing all possible sequences of moves in a game, with leaf nodes representing the outcomes.
- **Perceptron:** A fundamental unit in neural networks, consisting of weighted inputs, a summation node, and an activation function that produces an output.
- **Reinforcement Learning:** A type of machine learning where an agent learns to make decisions by interacting with an environment and receiving rewards or penalties.
- **Matrix:** A rectangular array of numbers, symbols, or expressions arranged in rows and columns.
- **Linear Transform:** A function that maps vectors from one vector space to another, satisfying $T(u+v) = T(u) + T(v)$ and $T(cu) = cT(u)$ for all vectors u, v and scalars c .
- **Matrix Multiplication:** An operation that combines two matrices to produce a third matrix, where the element at row i and column j of the resulting matrix is the dot product of row i of the first matrix and column j of the second matrix. The inner dimensions must match.
- **Matrix Inverse:** For a square matrix A , its inverse A^{-1} is a matrix such that $AA^{-1} = A^{-1}A = I$, where I is the identity matrix.
- **Matrix Transposition:** An operation that swaps the rows and columns of a matrix. The transpose of matrix A is denoted as A^T .
- **Derivative:** A measure of how a function changes in response to a small change in its input. It represents the instantaneous rate of change.
- **Partial Derivative:** A derivative of a function of multiple variables with respect to one of those variables, treating the others as constants.
- **Gradient:** A vector that points in the direction of the greatest rate of increase of a function at a given point. It is composed of the partial derivatives of the function with respect to each variable.
- **Sigmoid Function:** A mathematical function that maps any input value to an output value between 0 and 1. Often used as an activation function in neural networks.
- **Inductive Reasoning:** The process of deriving general principles from specific observations.

- **Linear Regression:** A machine learning algorithm used to model the relationship between a dependent variable and one or more independent variables by fitting a linear equation to observed data.
- **Model Evaluation:** The process of assessing the performance of a machine learning model.
- **Overfitting:** A phenomenon in machine learning where a model learns the training data too well, resulting in poor generalization to unseen data.
- **Representative Data:** Data that accurately reflects the characteristics of the problem domain being addressed.
- **Binary Classification:** A machine learning task where the goal is to classify data into one of two categories.
- **Decision Boundary:** A line or surface that separates different classes of data points in a classification problem.
- **0{-}1 Loss:** A loss function in binary classification that assigns a loss of 1 if the prediction is incorrect and 0 if it is correct.
- **Perceptron Loss:** A loss function used in the perceptron algorithm, defined as $\max\{0, -z\}$, where $z = y(w^T x)$.
- **Hinge Loss:** A loss function used in support vector machines (SVMs), defined as $\max\{0, 1 - z\}$, where $z = y(w^T x)$.
- **Logistic Loss:** A loss function used in logistic regression, defined as $\log(1 + e^{-z})$, where $z = y(w^T x)$.
- **Gradient Descent:** An iterative optimization algorithm used to find the minimum of a function by following the direction of the negative gradient.
- **Stochastic Gradient Descent:** A variant of gradient descent that updates the model parameters using a randomly selected subset of the data points.
- **Empirical Risk Minimizer:** A learning principle that aims to find the model that minimizes the average loss over the training data.

2 Problem Types

- **Understanding the Nature of Intelligence:** This problem explores the definition and characteristics of intelligence, both human and artificial.
- **Defining Artificial Intelligence:** This problem involves formulating a precise definition of artificial intelligence, considering various perspectives and historical contexts.
- **Passing the Turing Test:** This problem investigates whether passing the Turing Test is sufficient evidence for a machine possessing genuine intelligence, considering counterarguments like Searle's Chinese Room.
- **Search Space Complexity in Game Playing:** This problem addresses the computational challenges of searching game trees, particularly in complex games like chess and Go, highlighting the need for machine learning techniques beyond brute-force search.
- **Overcoming Limitations of Search Algorithms:** This problem focuses on the limitations of pure search algorithms in complex domains and the necessity of incorporating machine learning techniques to handle the vast search spaces.
- **Addressing AI Safety Concerns:** This problem explores the potential risks and vulnerabilities of AI systems, such as susceptibility to adversarial attacks and the presence of biases in algorithms.

- **Ethical Considerations in AI Development:** This problem examines the ethical implications of AI, including issues of bias, fairness, and accountability in AI systems.
- **Comparing Game Playing Algorithms:** This problem involves comparing and contrasting different game-playing algorithms, such as Minimax and Alpha-Beta pruning, and analyzing their strengths and weaknesses.
- **Understanding the Differences Between Game Types:** This problem explores the differences between various game types (e.g., chess vs. soccer) in terms of their complexity, dynamics, and the challenges they pose for AI development.
- **Similarities Between Seemingly Disparate AI Tasks:** This problem investigates the commonalities between different AI tasks, such as speech synthesis and game playing, in terms of algorithm design and problem-solving approaches.
- **The AI Stack and its Interconnectedness:** This problem explores the different layers and sub-fields within the broader field of Artificial Intelligence and their interconnectedness.
- **Understanding Perceptrons:** This problem involves understanding the structure and function of perceptrons, a fundamental building block of neural networks.
- **Reinforcement Learning in Game Playing:** This problem explores the application of reinforcement learning in game playing, using historical examples like Samuel's checkers program.
- **Analyzing AlphaGo's Success:** This problem involves analyzing the architecture and strategies employed by AlphaGo, a successful AI system for playing Go, and understanding its combination of deep learning and search techniques.
- **Representing Systems of Equations with Matrices:** This problem involves representing a system of linear equations using matrices, where the coefficients form the matrix A , the variables form the vector x , and the constants form the vector b , resulting in the equation $Ax = b$.
- **Solving Systems of Equations using Matrix Inverses:** This problem focuses on solving systems of linear equations represented in matrix form ($Ax = b$) by utilizing the inverse of the coefficient matrix (A^{-1}) to obtain the solution vector $x = A^{-1}b$.
- **Matrix Multiplication and Linear Transformations:** This problem explores matrix multiplication as a representation of linear transformations, where multiplying a matrix by a vector applies a transformation, and multiplying two matrices creates a composite transformation. The importance of matching inner dimensions for multiplication is also highlighted.
- **Matrix Addition, Subtraction, and Scalar Multiplication:** This problem covers the basic arithmetic operations on matrices, including element-wise addition and subtraction, and scalar multiplication, emphasizing the requirement of matching dimensions for addition and subtraction.
- **Matrix Inverses and their Properties:** This problem introduces the concept of matrix inverses, stating that only square matrices can have inverses and demonstrating that the product of a matrix and its inverse equals the identity matrix ($AA^{-1} = I$).
- **Matrix Transposition:** This problem introduces matrix transposition as a fundamental operation in linear algebra, where rows and columns of a matrix are interchanged to create the transpose (A^T).
- **Calculating the Derivative of the Sigmoid Function:** This problem involves calculating the derivative of the sigmoid function, a crucial step in many AI algorithms, particularly in neural networks. The derivation utilizes the chain rule and sum rule of calculus.
- **Parallelization of Matrix Multiplication:** This problem addresses the parallelization of matrix multiplication using domain decomposition, where the computation is divided among multiple processing cores to improve efficiency.

- **Linear Regression:** Finding the line that best fits a set of data points, minimizing the sum of squared distances between the points and the line.
- **Multiple Linear Regression:** Extending linear regression to include multiple independent variables to predict a dependent variable.
- **Curve Fitting:** Fitting a curve to a set of data points, potentially using polynomial functions of varying degrees.
- **Binary Classification:** Classifying data points into two distinct categories using a learned function.
- **Choosing a Loss Function:** Selecting an appropriate loss function (e.g., 0-1 loss, perceptron loss, hinge loss, logistic loss) to measure the error of a classification model.
- **Minimizing Loss using Gradient Descent:** Employing gradient descent or stochastic gradient descent to iteratively update model parameters and minimize the chosen loss function.

3 Algorithm Solutions

- **Minimax Search:** A recursive algorithm for two-player zero-sum games that explores the game tree to find the best move for the current player, assuming the opponent also plays optimally. It alternates between maximizing the score for the current player and minimizing the score for the opponent.
- **Alpha{-}Beta Pruning:** An optimization technique for Minimax search that reduces the search space by avoiding the evaluation of branches that cannot affect the optimal decision. It uses alpha and beta values to prune branches.
- **Matrix Addition:** $A + B = C$, where $c_{ij} = a_{ij} + b_{ij}$
- **Matrix Scalar Multiplication:** $k \cdot A = B$, where $b_{ij} = k \cdot a_{ij}$
- **Matrix Multiplication:** $C_{ij} = \sum_{k=1}^n a_{ik} b_{kj}$
- **Matrix Transposition:** Swapping rows and columns of a matrix A to obtain A^T .
- **Solving Linear Equations with Invertible Matrices:** $Ax = b \implies x = A^{-1}b$
- **Derivative of Sigmoid Function:** $\frac{d}{dx}\sigma(x) = \frac{e^{-x}}{(1 + e^{-x})^2}$
- **Perceptron Algorithm:** Iteratively update the weight vector w based on misclassified data points. If $\text{sign}(w^T x_n) \neq y_n$, then $w \leftarrow w + y_n x_n$.